

Datagen: LLM-generated near-miss negatives do not transfer to real documents

Results

2,000 adversarial near-miss negatives were generated with `claude -p` across ten classes, verified, added to the document gate's training set at four weights, and measured on the sealed real-world corpora. **The result is negative and the mechanism is now certain.**

generated-data weight	sealed real AUC	datax AUC	govdocs2 AUC	sealed real recall	held-out generated fire rate
0.0 (<i>baseline</i>)	0.8453	0.8504	0.8434	0.4979	8.2%
1.0	0.8437	0.8471	0.8432	0.4786	0.0%
4.0	0.8413	0.8388	0.8446	0.4749	0.0%
12.0	0.8461	0.8461	0.8469	0.4738	0.0%

(recall at matched specificity 0.95; AUC is threshold-free. 1,556 generated documents added to 432,903 fit rows; 392 held out and never trained on.)

The gate learns the generated documents **perfectly** — it stops firing on 100% of held-out generated near-misses, which it has never seen — and the sealed real-world number does not move at all. Recall on real documents gets slightly worse.

The mechanism, measured

A classifier trained to answer "is this document machine-generated?" separates the 1,948 generated negatives from 5,152 real negatives at a **held-out AUC of 1.0000**.

That is the whole explanation. Style is a free, perfectly predictive feature. A gate can satisfy every generated row by detecting Claude's prose, so gradient descent has no pressure whatsoever to learn the thing the corpus was built to teach — that a form with no values, a schema with no rows, or a policy *about* identifiers is not a document containing identifiers. The corpus taught a shortcut, and the shortcut was cheaper than the concept.

What the verification caught

Of 2,000 documents, **52 were rejected (2.6%)**:

reason	n
tell:disclaimer	13
identifier:iban	13
identifier:ipv4	10
identifier:street_address	9

reason	n
identifier:ssn	5
identifier:card	5
identifier:email	2

The 39 identifier hits are **mislabelled negatives** — documents asserting `doc_has_pii = False` that contain a social security number, a card number or a street address. Each would have taught the gate to stay shut on a document that does carry PII, which is the precise failure this project exists to prevent. The 13 disclaimers would have handed the model a single sentence to memorise.

The surviving 1,948 were genuinely adversarial: the champion gate wrongly fires on **23.3%** of them, against **15.5%** of real govdocs2 negatives. They were harder than the real thing — and still useless, because they were separable.

TL;DR

- **Generated near-miss negatives do not work.** Sealed real AUC 0.8453 → 0.8461 across four weights: flat. Real-document recall slightly worse.
- **The gate learns them completely** — held-out generated fire rate 8.2% → 0.0%. Every training-side metric improves. None of it transfers.
- **Cause is measured, not inferred:** generated vs real negatives separate at **AUC 1.0000**. The model learns "machine-generated" instead of the concept.
- **The verification gate earned its place:** it rejected 39 documents carrying real identifiers under a negative label, plus 13 shared tells.
- **This does not invalidate the diagnosis.** Real-world negatives being harder is still the binding constraint; synthesising them is what fails.
- **The next lever is real documents, not generated ones** — active learning on an unlabelled real corpus (below).

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Author	Ryan Lence
Project	projects/pii-head-to-head-v1
Run ID	H1 (datagen follow-up)
Scope	document gate only; tag heads untouched
Outcome	negative result, mechanism identified, direction changed

Artifacts

Path	What
datagen/2000_26082521_claude/	2,000 documents, manifest.json, manifest_verified.json, verification.json
probe/gate_augment.json	four gate refits: sealed AUC, recall, held-out fire rate
training/h2h_datagen.py	the generator
training/h2h_datagen_verify.py	the verification gate
training/h2h_gate_augment.py	the augmentation measurement
reports/26-08-25-1_gate-diagnosis.md	the diagnosis this follows from

Model scoreboard

Model	Dataset	F1@5	F1@3	F1@1	Macro F1	Docs/sec	ms/doc
fusion-1k	10360_betterdataai_ner_silver_eval_10.36k				0.208	1360.8	0.7
fusion-1k	10626_ai4privacy_pii_masking_eval_10.63k				0.424	1360.8	0.7
fusion-1k	20000_pii_holdout_20.00k				0.477	1360.8	0.7
fusion-1k	30000_pii2_eval_25.15k				0.413	1360.8	0.7
fusion-1k	38937_openpii_pii_eval_38.94k				0.510	1360.8	0.7
fusion-1k	4000_datax-dualjudge-evalset-1.32k					1360.8	0.7
fusion-1k	5617_nemotron_eval_5.36k					1360.8	0.7
fusion-1k	6589_govdocs2-dualjudge-eval20-3.53k					1360.8	0.7
steady-cascade	10360_betterdataai_ner_silver_eval_10.36k				0.266	314.1	3.2
steady-cascade	10626_ai4privacy_pii_masking_eval_10.63k				0.729	314.1	3.2
steady-cascade	20000_pii_holdout_20.00k				0.709	314.1	3.2
steady-cascade	30000_pii2_eval_25.15k				0.757	314.1	3.2
steady-cascade	38937_openpii_pii_eval_38.94k				0.657	314.1	3.2
steady-cascade	4000_datax-dualjudge-evalset-1.32k					314.1	3.2
steady-cascade	5617_nemotron_eval_5.36k					314.1	3.2
steady-cascade	6589_govdocs2-dualjudge-eval20-3.53k					314.1	3.2
fusion-12k	10360_betterdataai_ner_silver_eval_10.36k				0.208	270.3	3.7
fusion-12k	10626_ai4privacy_pii_masking_eval_10.63k				0.425	270.3	3.7
fusion-12k	20000_pii_holdout_20.00k				0.485	270.3	3.7

Model	Dataset	F1@ 5	F1@ 3	F1@ 1	Macro F1	Docs/s ec	ms/d oc
fusion-12k	30000_pii2_eval_25.15k				0.390	270.3	3.7
fusion-12k	38937_openpii_pii_eval_38.94k				0.511	270.3	3.7
fusion-12k	4000_datax-dualjudge-evalset-1.32k					270.3	3.7
fusion-12k	5617_nemotron_eval_5.36k					270.3	3.7
fusion-12k	6589_govdocs2-dualjudge-eval20-3.53k					270.3	3.7

Measured 2026-08-25 · n=10,360, n=10,626, n=20,000, n=30,000, n=38,937, n=4,000, n=5,617, n=6,589.